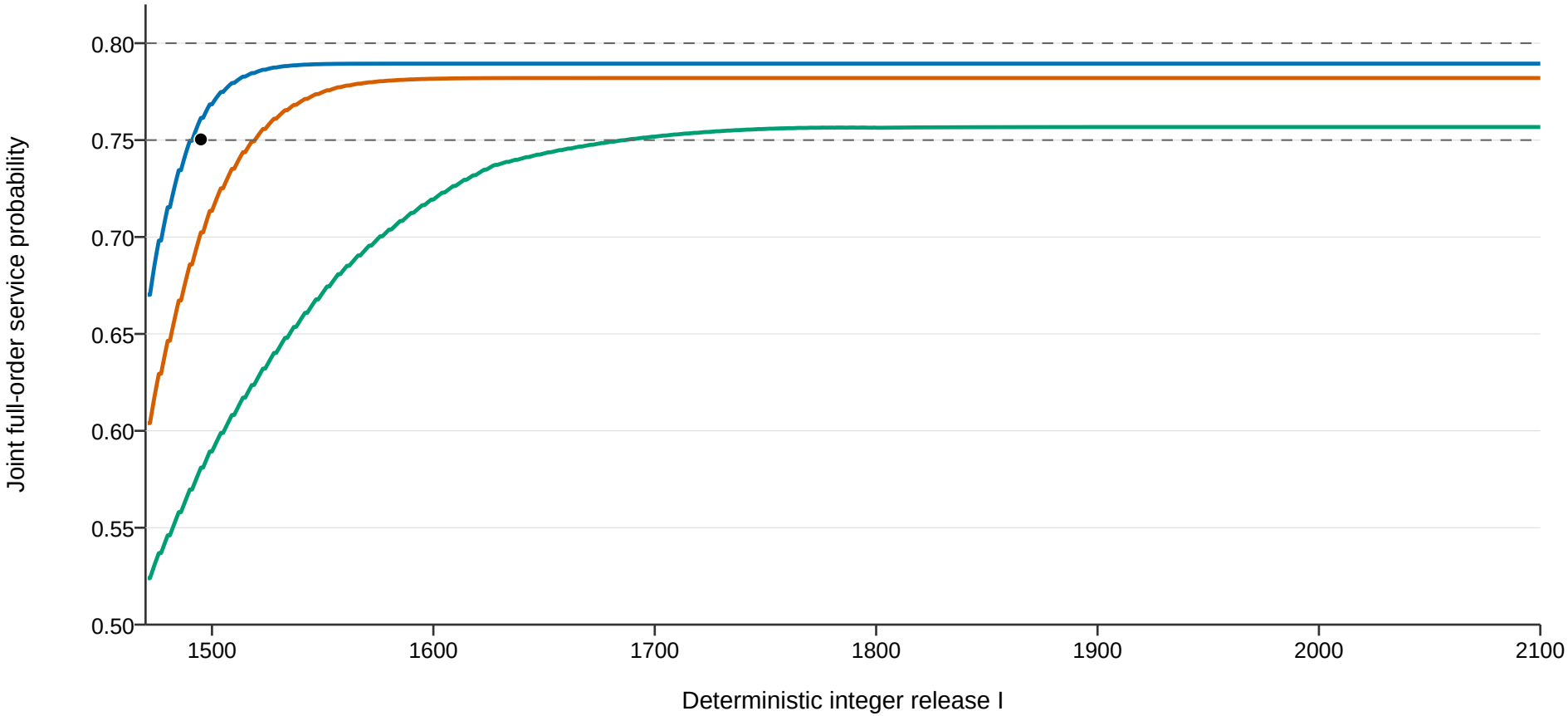
Complete-order service under a candidate final-station repair route

D=(750,150); normal-source class probabilities; eta=0.97; setup workload=0



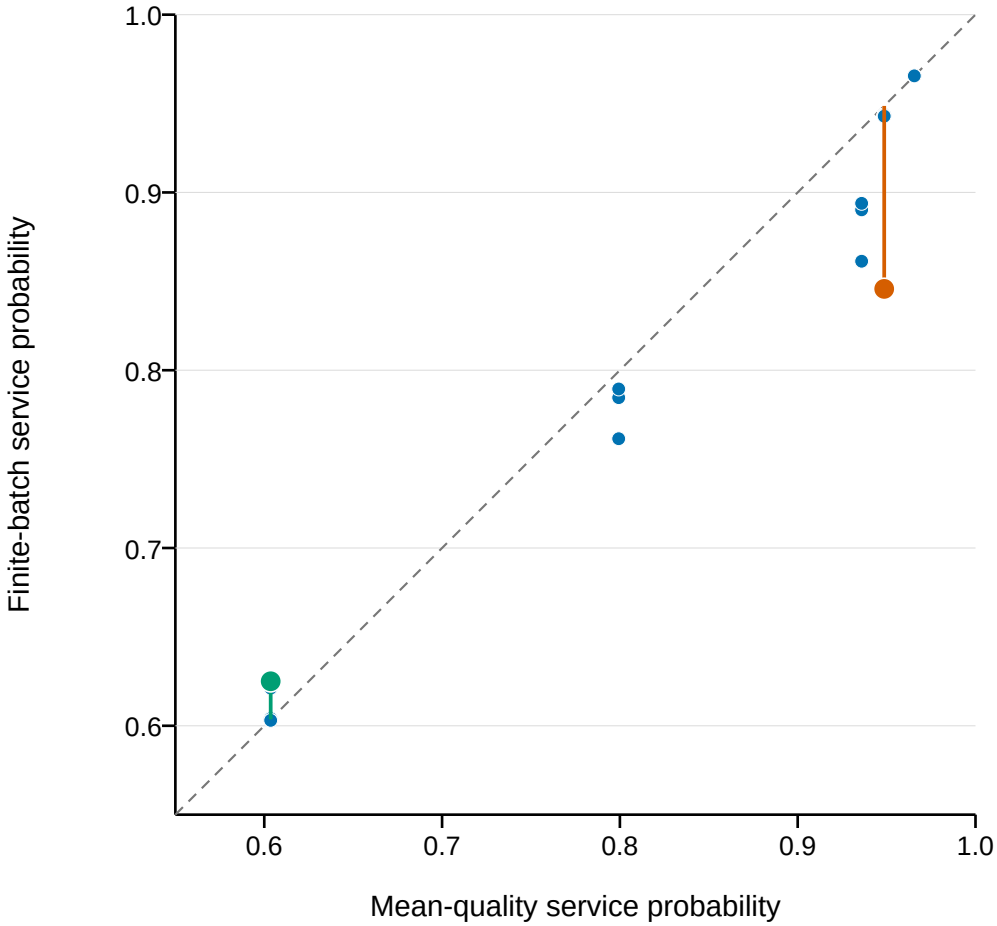
Black dot: selected iid setup policy, I=1495, service=0.75035 (F=2000). Dashed: 0.75 and 0.80 targets.

Hypothetical quality laws and repair route; fixed forward flow. Integer-grid segments guide the eye.

Source: saved rounds63/67. Curves show service capability; setup fees affect profit, not the capability ceiling.

Mean-quality plug-in estimates can overstate or understate service

Matched output floor and integer returns; eta=0.97; S=0; maximal returns in every profile



Largest overestimate in this subset

D=650/250, I=1316.2023

Mean 94.87% vs finite 84.57%

Difference: +10.29 percentage points

Largest underestimate in this subset

D=750/337, I=1699

Mean 60.36% vs finite 62.50%

Difference: -2.14 percentage points

34 predefined inputs, 15 source demand pairs.

Dashed line: estimates agree.

Points below line: plug-in overestimates.

Same extended model and repair rule throughout.

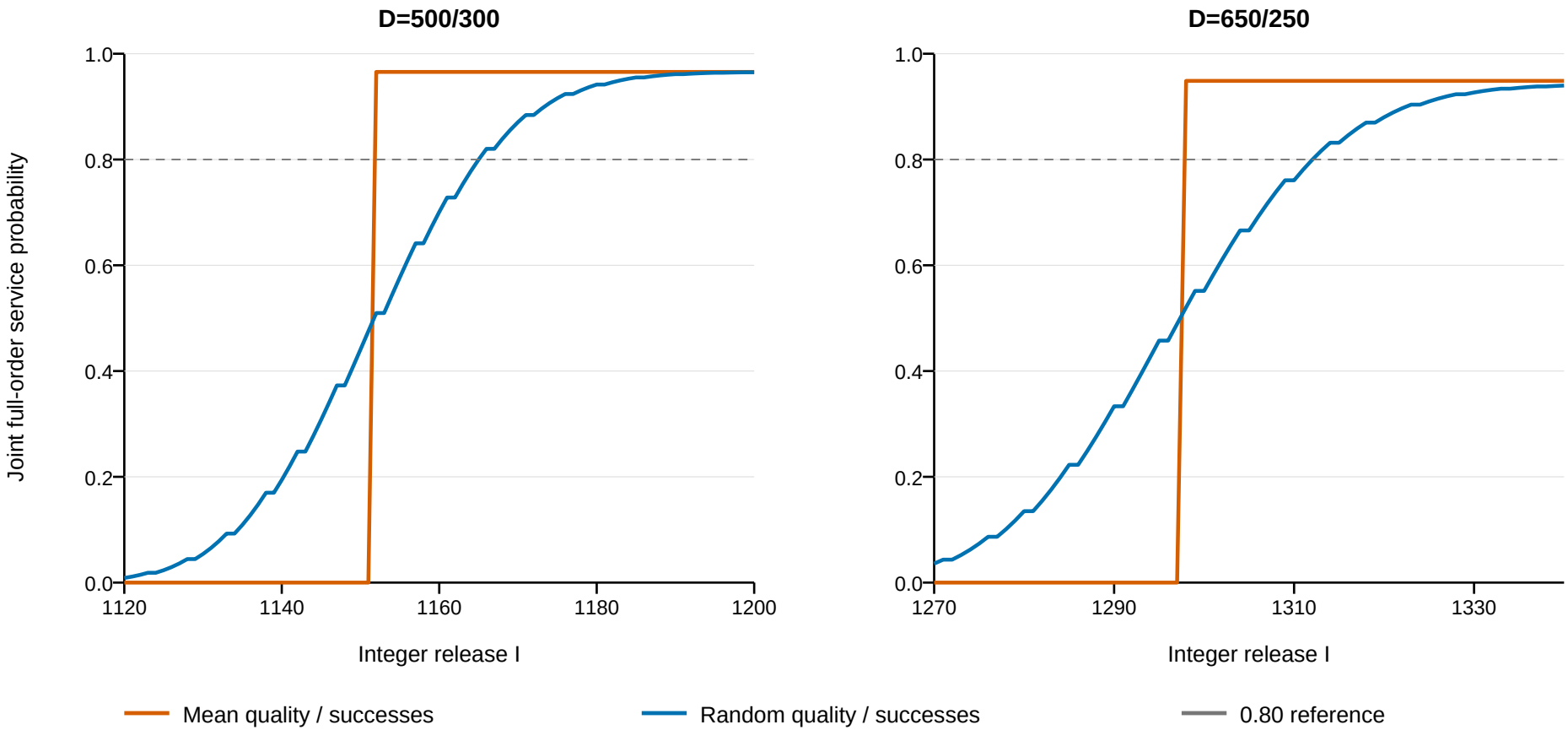
Source's own fixed-fraction assumptions kept separate.

Exact finite-law model comparison; no Monte Carlo error bars. Always return, not economic startup policies.

Source: saved round76. Inputs fixed by mean formula, first maximum-service integer, and economic input anchors.

A mean-quality threshold can hide finite-batch service uncertainty

Same output floor and integer return rule; mean counts/successes vs random counts/successes



D=500/300, I=1152: mean 0.96559 vs finite 0.50967 (+45.59 percentage points).

D=650/250, I=1297: mean 0 vs finite 0.48917 (-48.92 percentage points).

Source: round79 full grid (15 orders x 2101 inputs); 151 false >=0.80 claims. Lines guide integer points only.

Hypothetical extended model; no factory validation and no claim that the original fixed-fraction model is erroneous.